

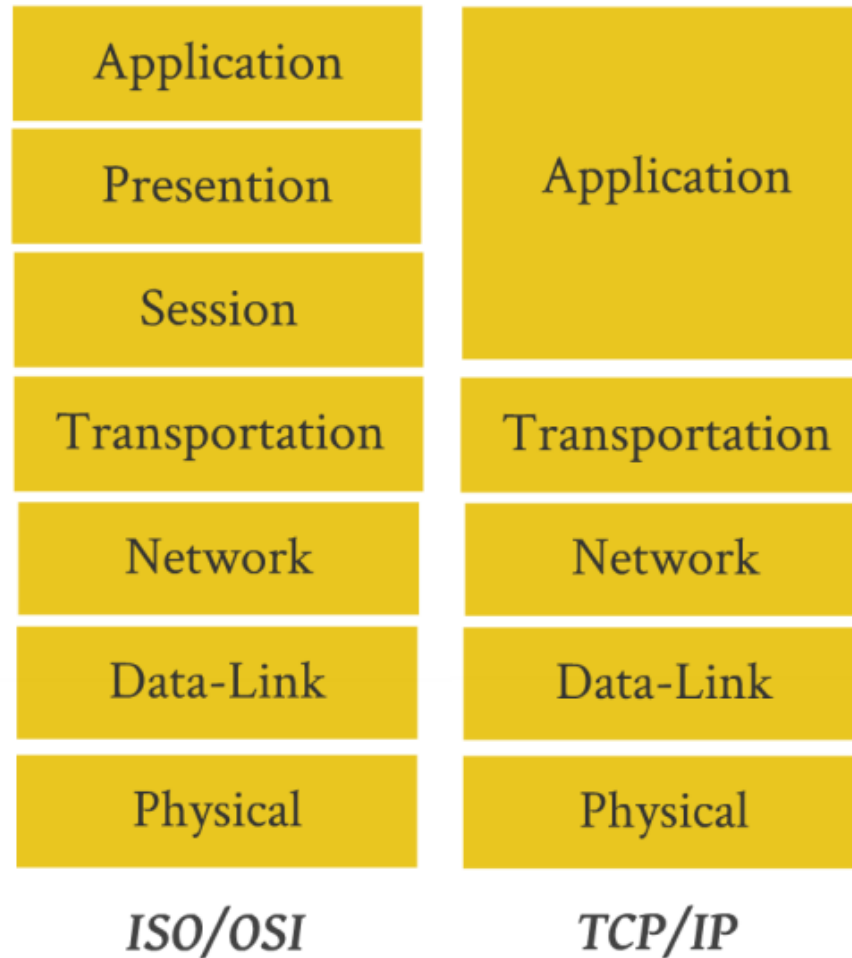
IBS: Bezpečnost a počítačové sítě

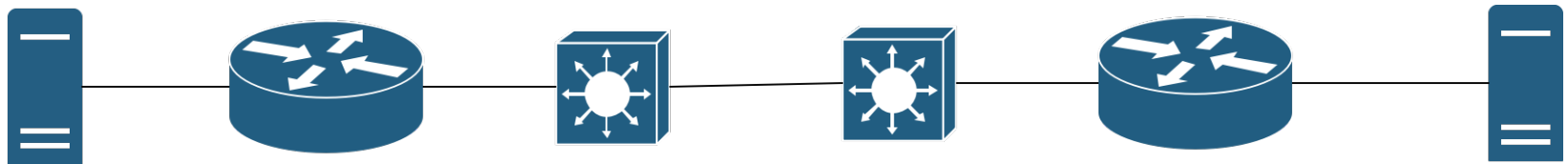
Overlay networks

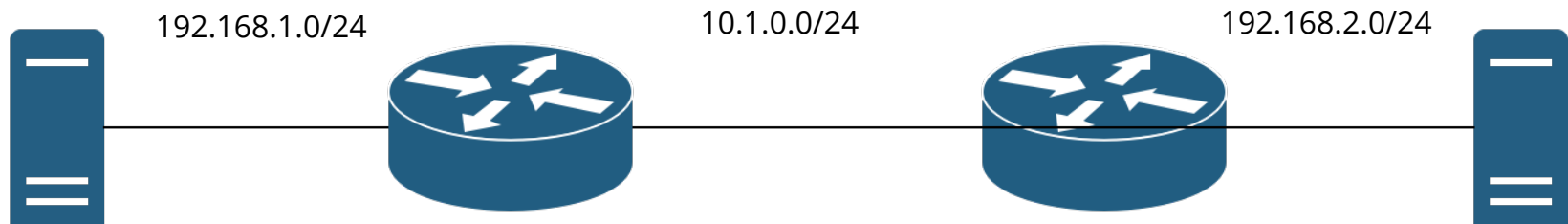
Matěj Grégr, mgregr@fit.vut.cz

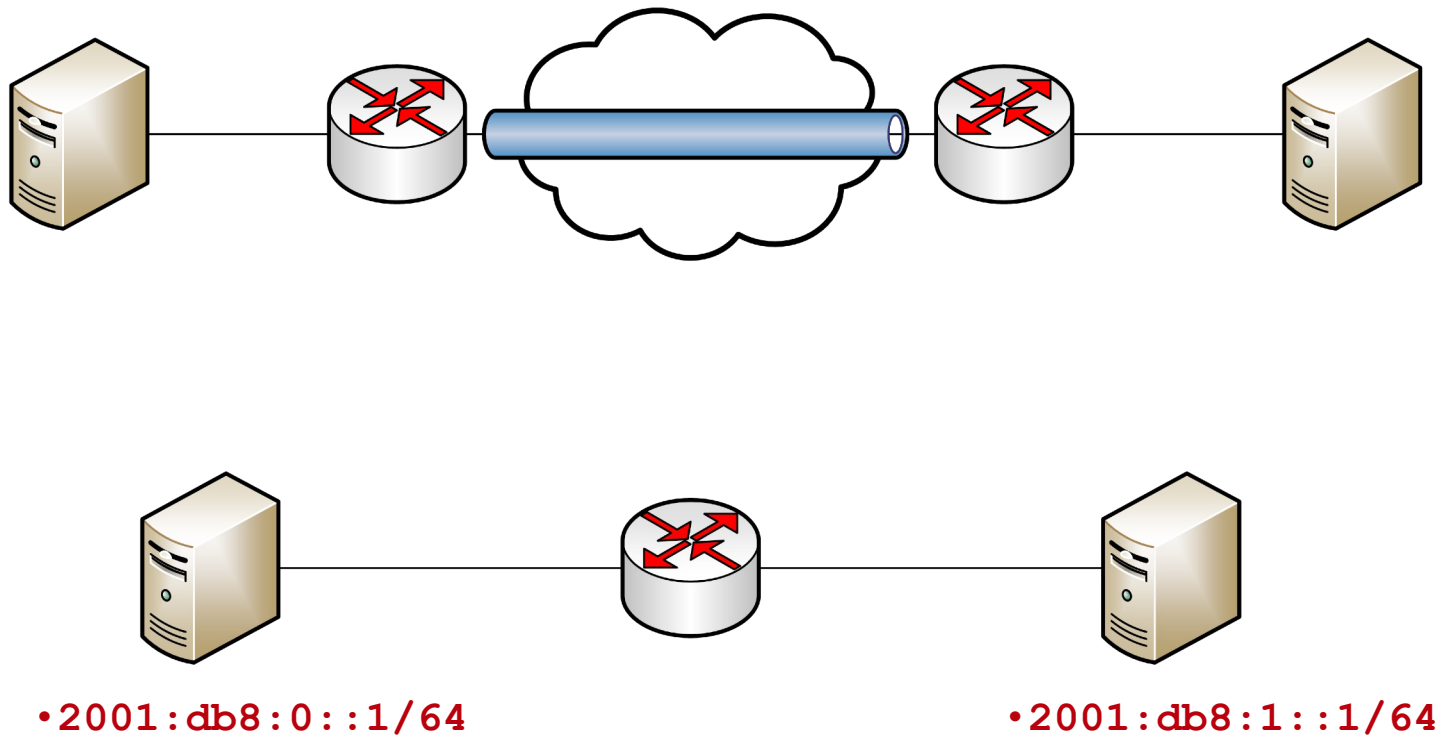


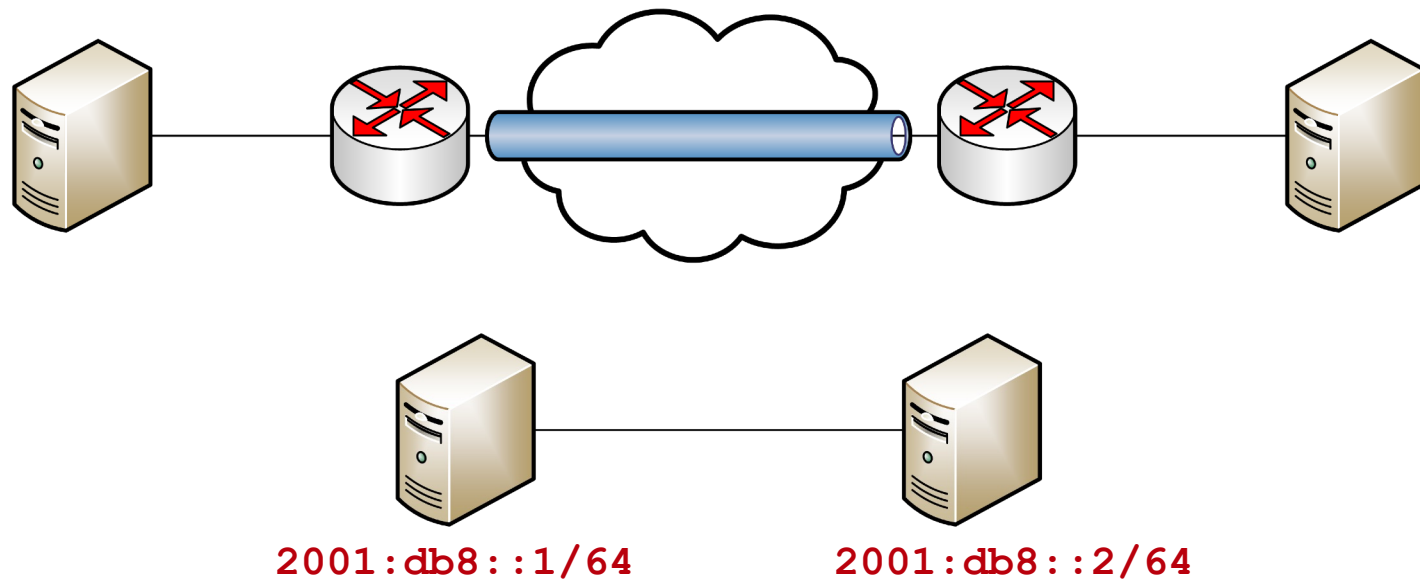
- L2-L3
 - VPN/MPLS
- Application overlays
 - CDN
 - Tor/I2P
 - Bittorrent...







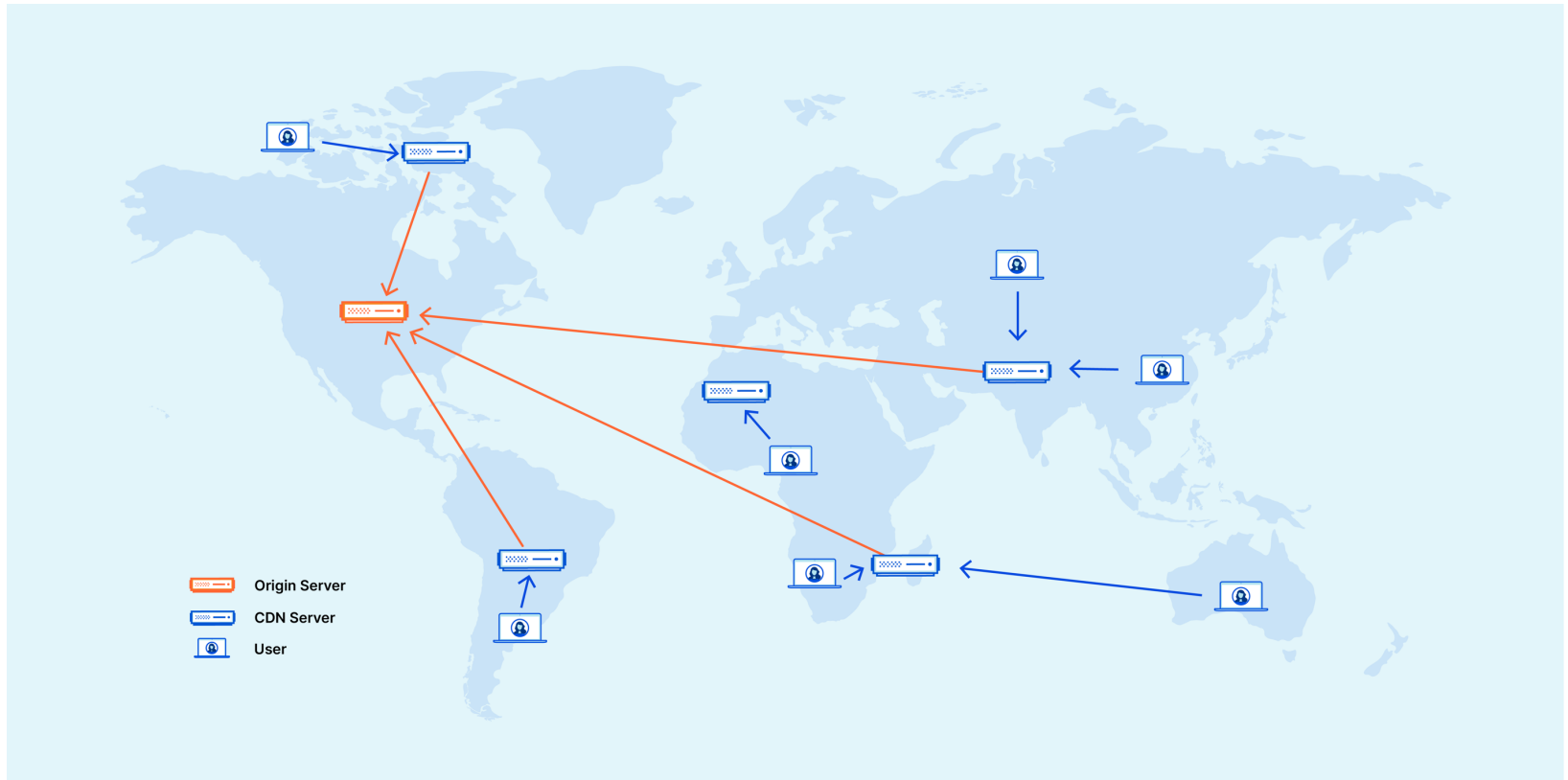




- L3 VPN
 - MPLS, IPSec, GRE, PPPoE, Openvpn, Wireguard ...
- L2 VPN
 - MPLS (VPLS), L2TP, VxLAN, Openvpn (tap - deprecated)

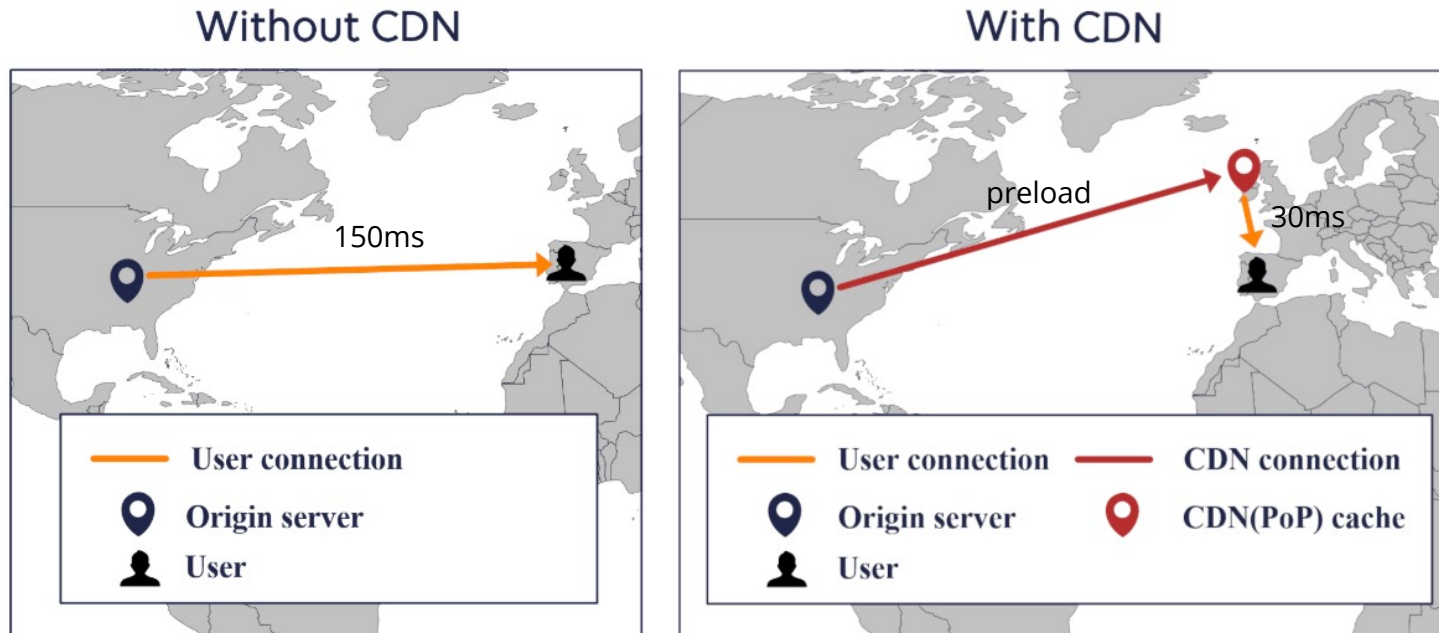
CDN

- How to stream large files (e.g., video) in real time?
 - replicate content at hundreds of servers throughout Internet
 - content downloaded to CDN servers ahead of time
 - placing content “close” to user avoids impairments (loss, delay) of sending content over long paths
 - CDN server typically in edge/access network
- Typical use-cases
 - Decreasing latency
 - Added fallback scenarios
 - Increasing security
- CloudFlare, Amazon CloudFront, Google, Fastly, Akamai, CDN77, KeyCDN
- Custom CDN solutions : <https://www.root.cz/serialy/jak-se-stavi-cdn/>

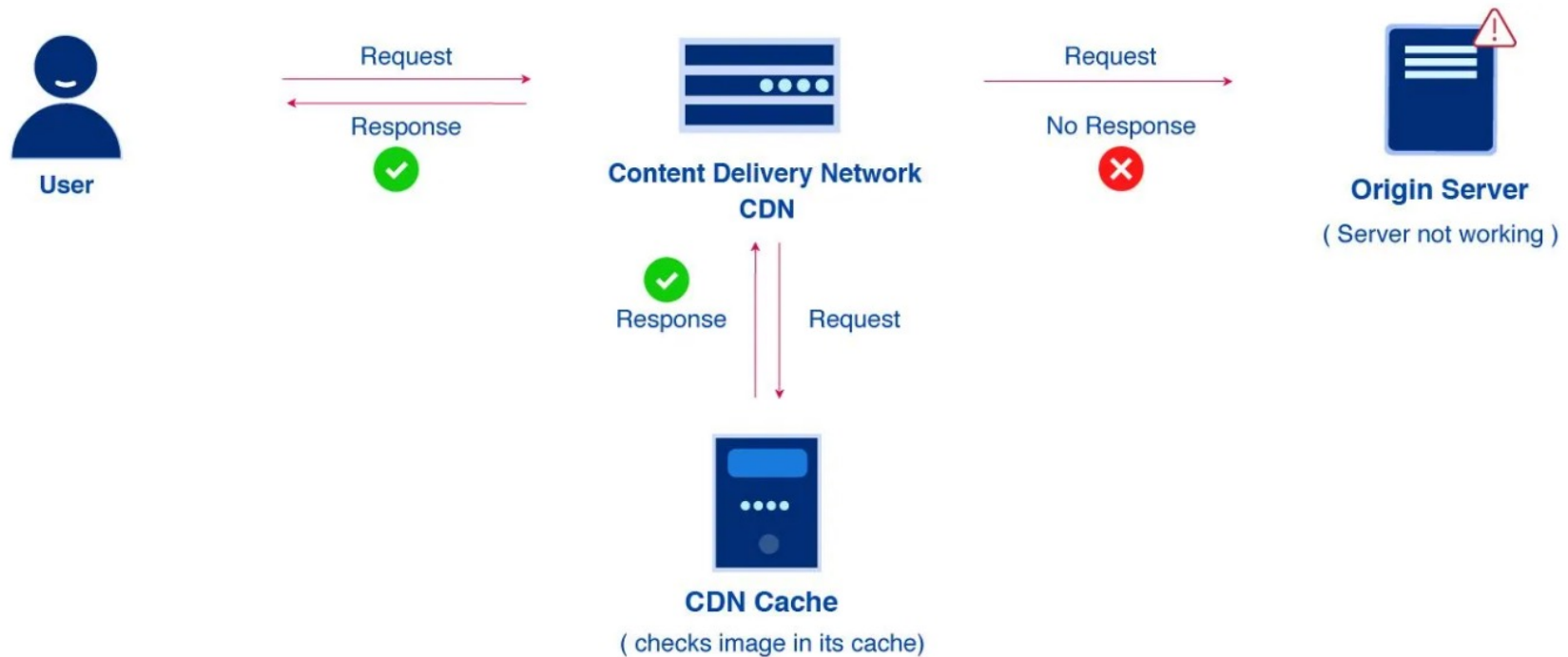


Source: <https://www.cloudflare.com/learning/cdn/what-is-a-cdn/>

- Typically for selective content only



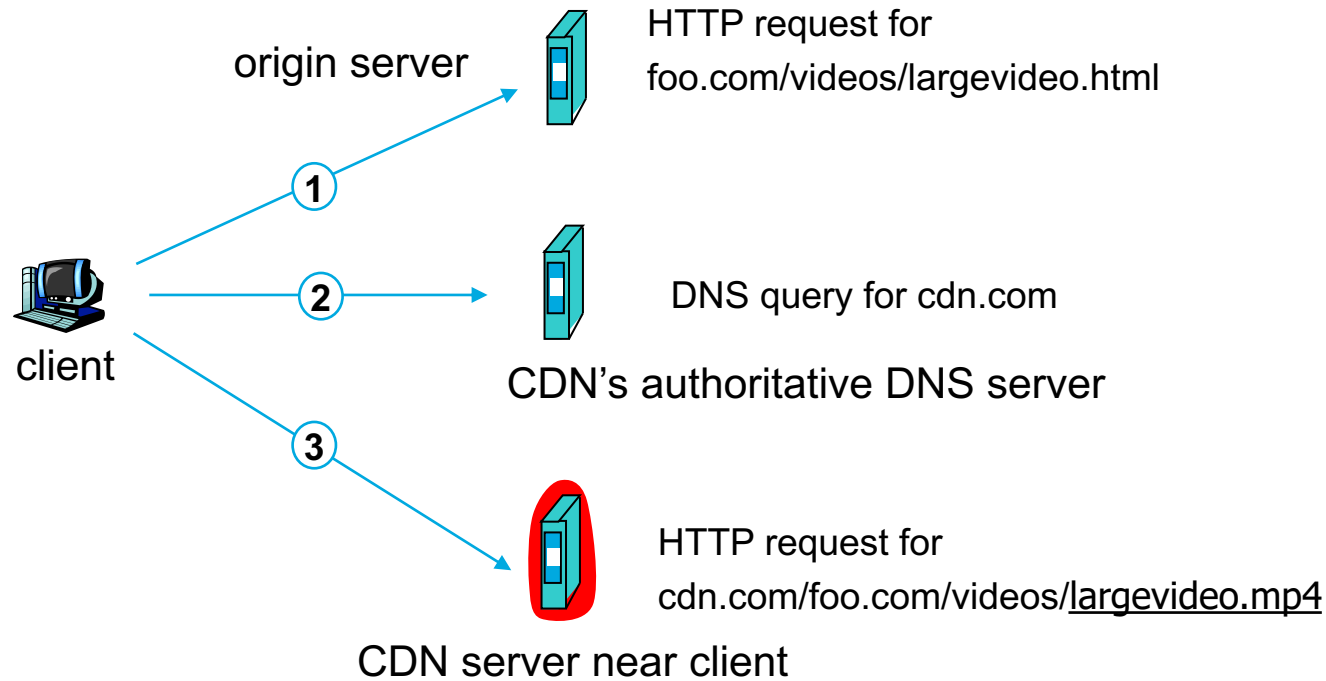
Source: <https://www.stormit.cloud/blog/what-is-an-origin-server/>



The CDN delivers content from its own cache while the origin server is down.



CDN blocks the request from a malicious user while serving good requests as usual



origin server

- distributes HTML
- `http://foo.com/videos/largevideo.mp4` → `http://cdn.com/foo.com/videos/largevideo.mp4`

CDN company (cdn.com)

- distributes gif files
- uses its authoritative DNS server to route redirect request

routing requests

- CDN creates a “map”, indicating distances from leaf ISPs and CDN nodes
- when query arrives at authoritative DNS server:
 - server determines ISP from which query originates
 - uses “map” to determine best CDN server
- CDN nodes create application-layer overlay network

P2P overlay networks

- Client-server
 - Many clients, 1 (or more) server(s)
 - Web servers, DNS, file downloads, video streaming
- Problems
 - Scalability: how many users can a server support?
 - What happens when user traffic overload servers?
 - Limited resources (bandwidth, CPU, storage)
 - Reliability: if # of servers is small, what happens when they break, fail, get disconnected, are mismanaged by humans?
 - Efficiency: if your users are spread across the entire globe, how do you make sure you answer their requests quickly?

- A simple idea
 - Users bring their own resources to the table
 - A cooperative model: clients = peers = servers
- The benefits
 - Scalability: # of “servers” grows with users
 - Reliability: load spread across many peers
 - Probability of them all failing is **very** low...
 - Efficiency: peers are distributed
 - Peers can try and get service from nearby peers

- Protocol: how do peers talk to each other?
- Service/data location: how do peers know who to talk to?
- Routing: how do you get a message from peer *A* to peer *B*?
- New reliability challenges
 - Network reachability, i.e., dealing with NATs
 - Dealing with churn, i.e., short peer uptimes
- Security
 - Malicious peers and cheating
 - The Sybil attack

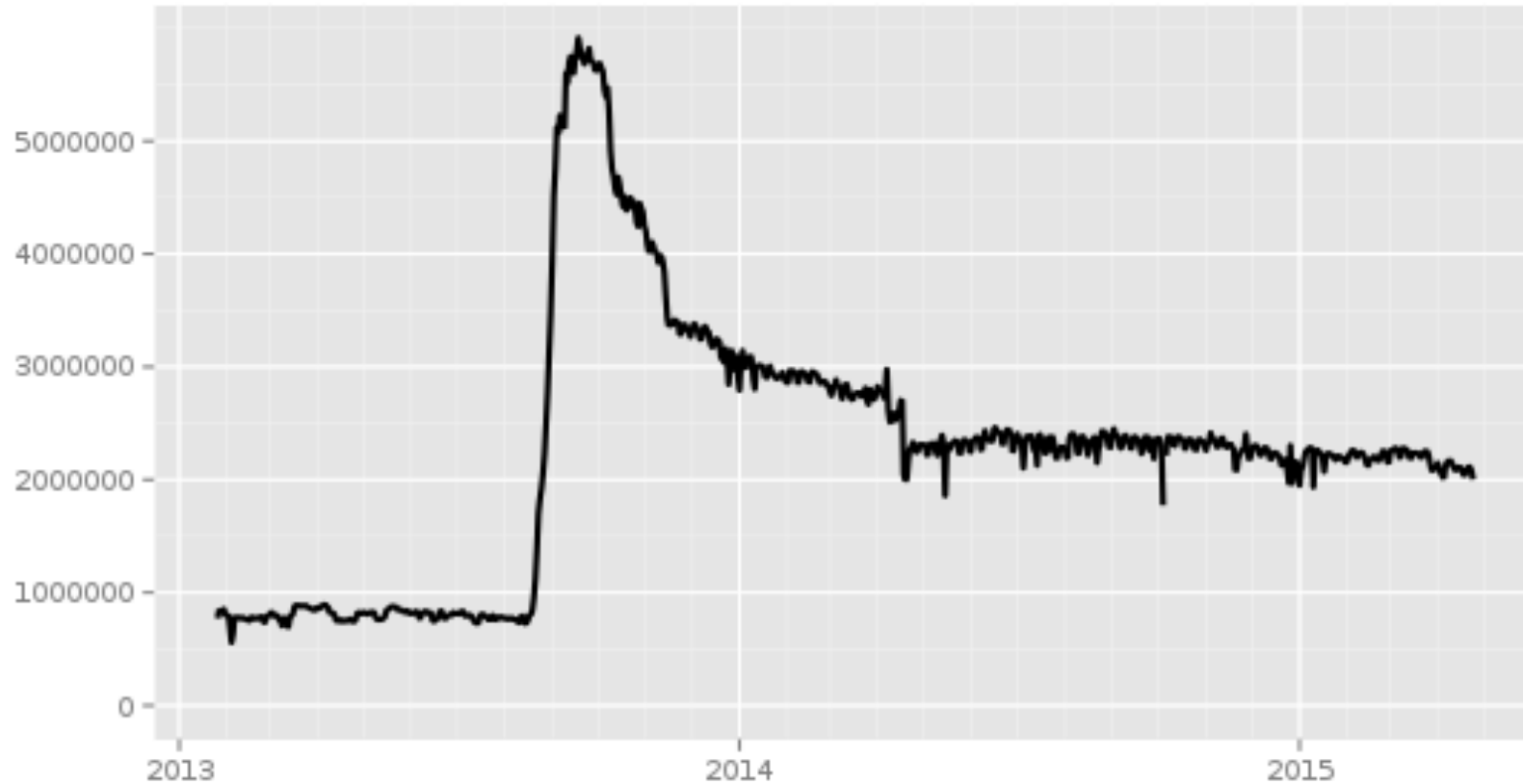
- Centralized Approach
 - The original: Napster
 - Single points of failure
- Unstructured P2P
 - No central server, peers only connect to each other
 - Queries sent as controlled flood
 - Later systems are hierarchical for performance reasons
- Limitations
 - Bootstrapping: how to join without central knowledge?
 - Floods of traffic = high network overhead
 - Probabilistic: can only search a small portion of the system

- Gnutella
 - One of the first massively popular unstructured P2P application
- Skype: P2P VoIP
 - Microsoft purchased Skype and centralized their architecture
- Hierarchical P2P Networks
 - FastTrack network
 - BitTorrent (Not a full-fledged P2P system – lack of file search)
- Bitcoin + several other cryptocurrencies

Tor

- Advanced mix network:
 - **Guards**: enhance source anonymity
 - **Relays**: mix proxy
 - **Hidden services**: servers accesible only through Tor (darknet)
- ~7000 Tor relays
 - Run by volunteers, universities, organisations
 - It's expected that some run by „intelligence agencies“
- 2 – 3 mil. users

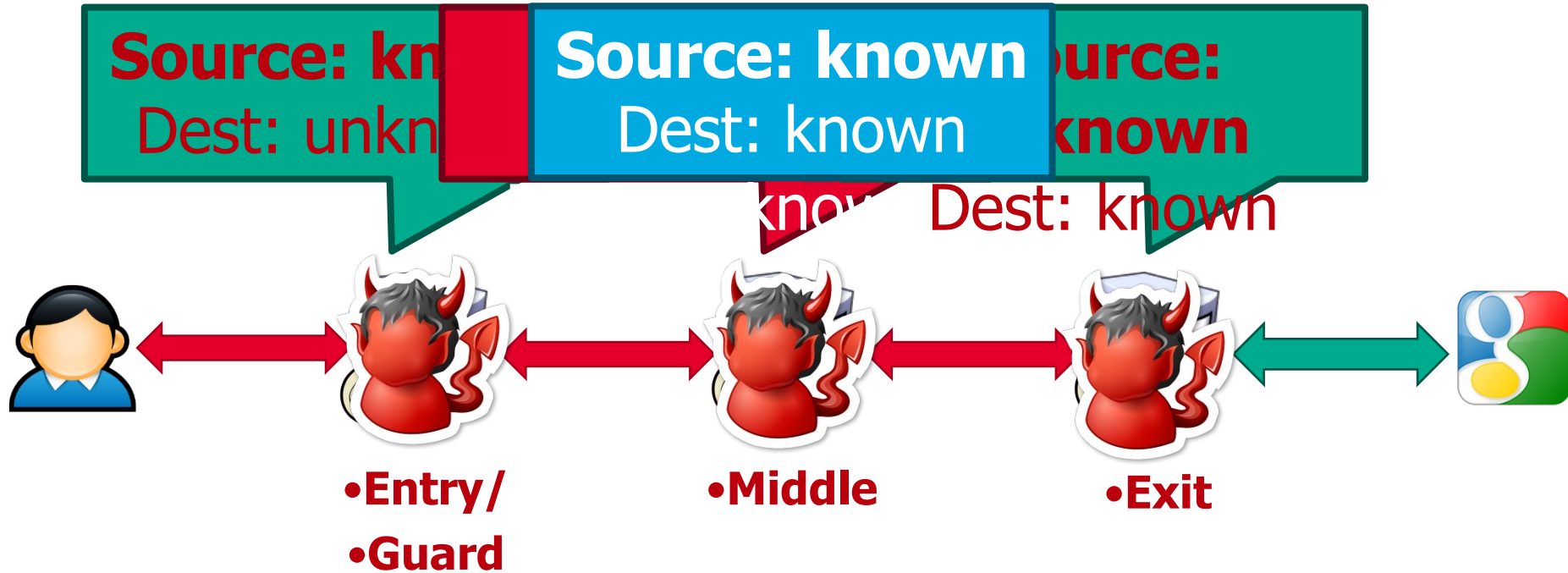
Directly connecting users



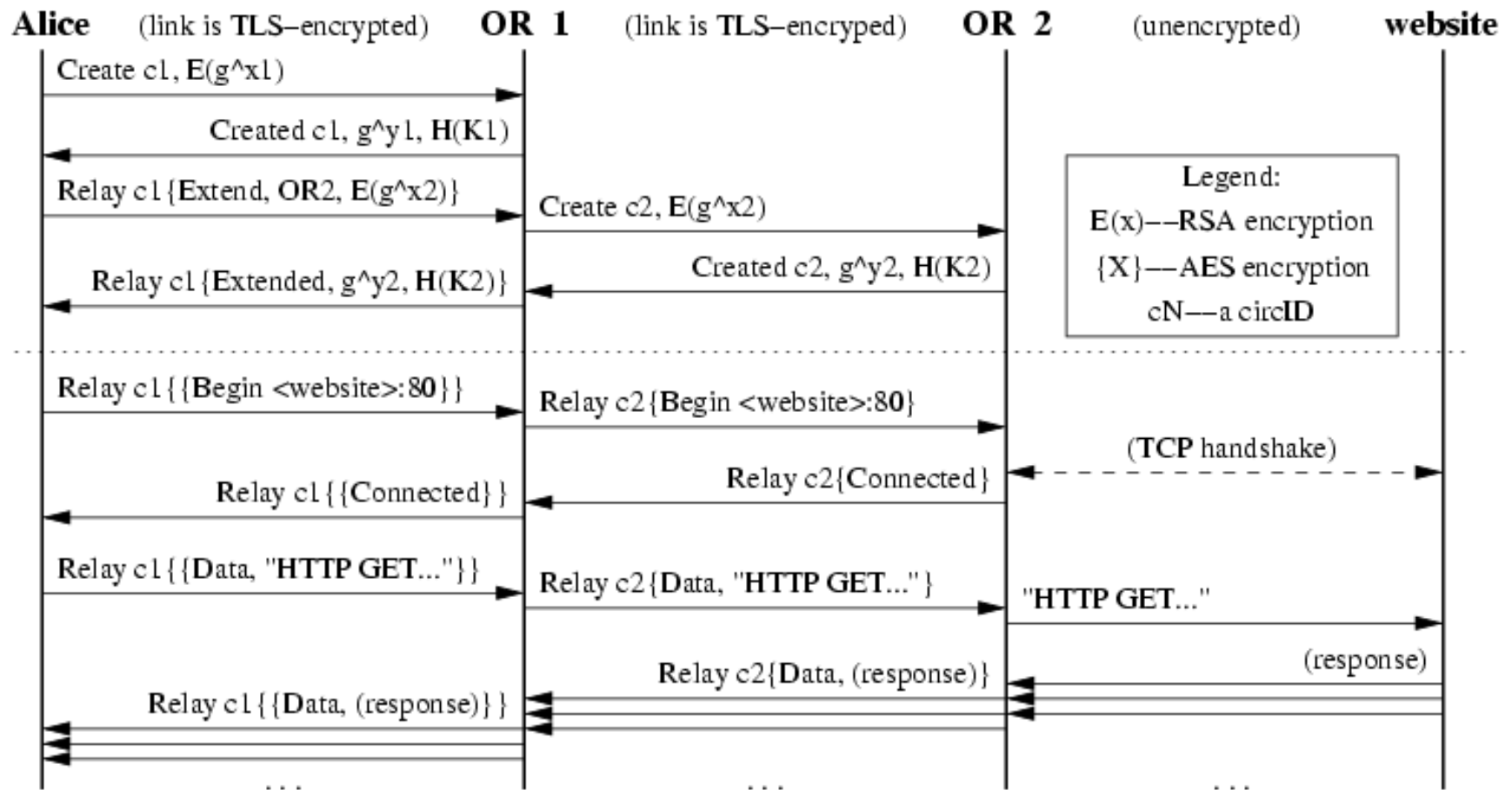
The Tor Project - <https://metrics.torproject.org/>

S“

- Tor client acts as a SOCKS proxy
 - Creates and manages **circuits**
- Support for all application that can connect via SOCKS proxy
- How to find a Tor relay?
 - Tor Consensus File
 - Hosted by **directory** servers
 - IP addresses of all known relays
 - IP address, uptime, measured bandwidth, etc.
- Packets are divided to cells 512 bajtů
 - Decrease speed, increase anonymity
- <https://gitweb.torproject.org/torspec.git/blob/HEAD:/tor-spec.txt>



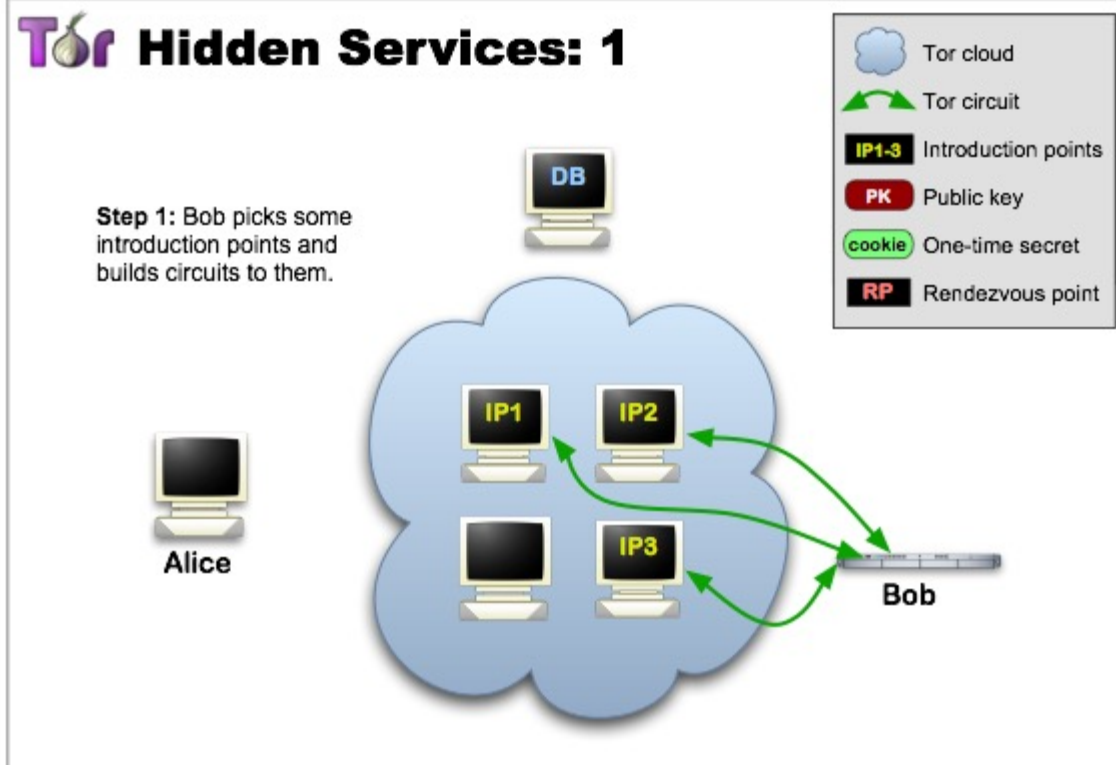
- Number of relays can be selected

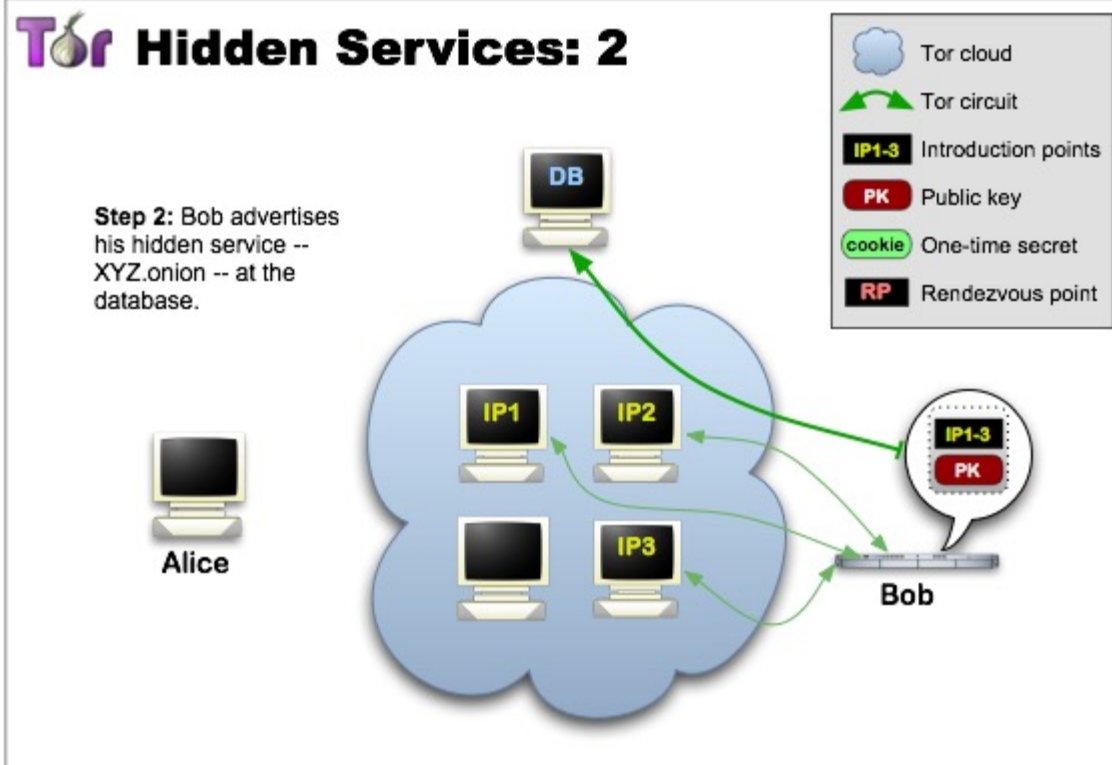


- Guard relays protect against attacker that acts as a entry relay
 - Tor picks 3 guard relays and use them for some time
 - The latest Tor version choose only one guard relay.
Why?
- Guard relay:
 - Large uptime
 - Bandwidth

- Assumption:
 - N : number of relays
 - M : number of relays controlled by an attacker
- Attacker's goal – control entry and exit relay
 - $\sim M/N$ probability for entry relay
 - $\sim (M-1)/(N-1)$ probability for exit relay
 - $\sim (M/N)^2$ probability for one circuit
- Client creates new circuits
 - Attacker chance increase during long time period

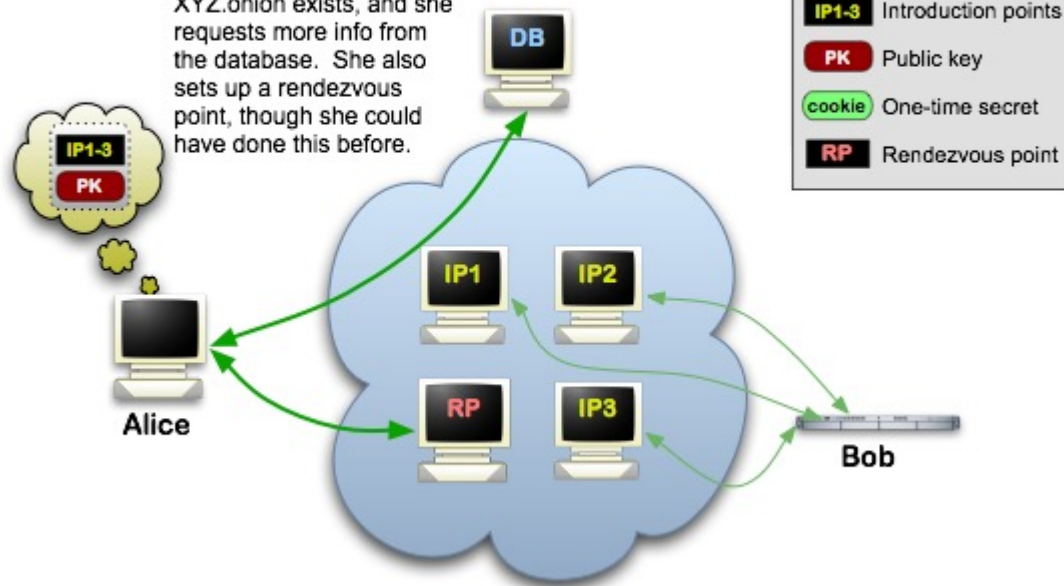
- Web services can be easily monitored
- Tor Hidden Services
 - Allows run an anonymous server without exposing its IP address or DNS name
- Services
 - Tor Mail, Tor Char
 - DuckDuckGo
 - Wikileaks
 - The Pirate Bay, Silk Road ...





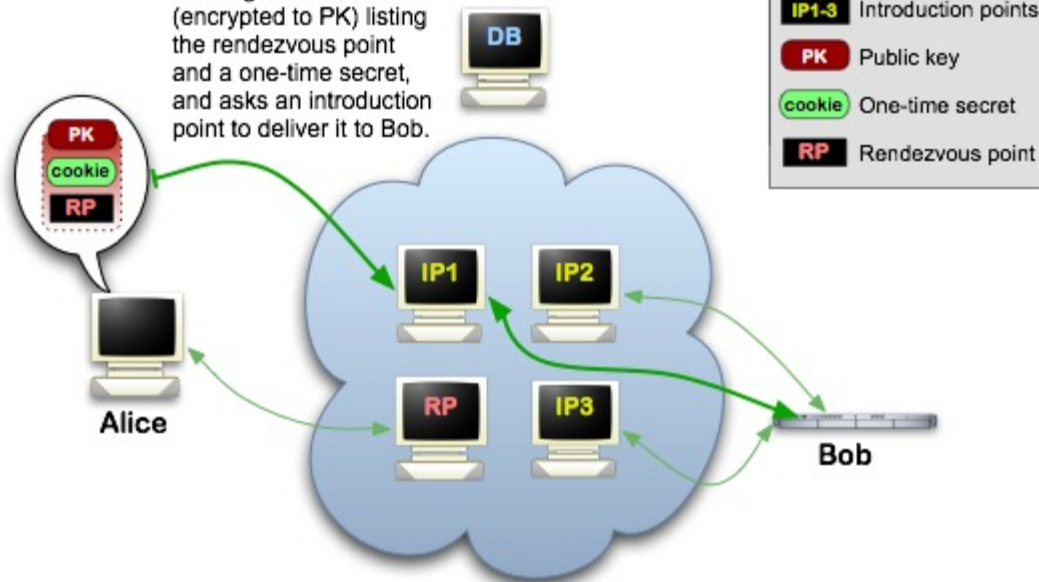
Tor Hidden Services: 3

Step 3: Alice hears that XYZ.onion exists, and she requests more info from the database. She also sets up a rendezvous point, though she could have done this before.



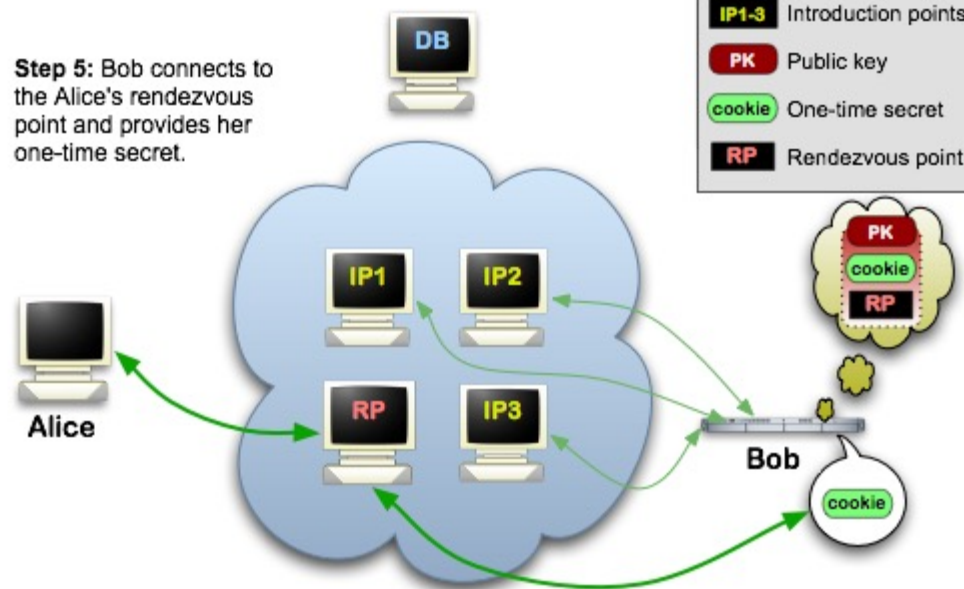
Tor Hidden Services: 4

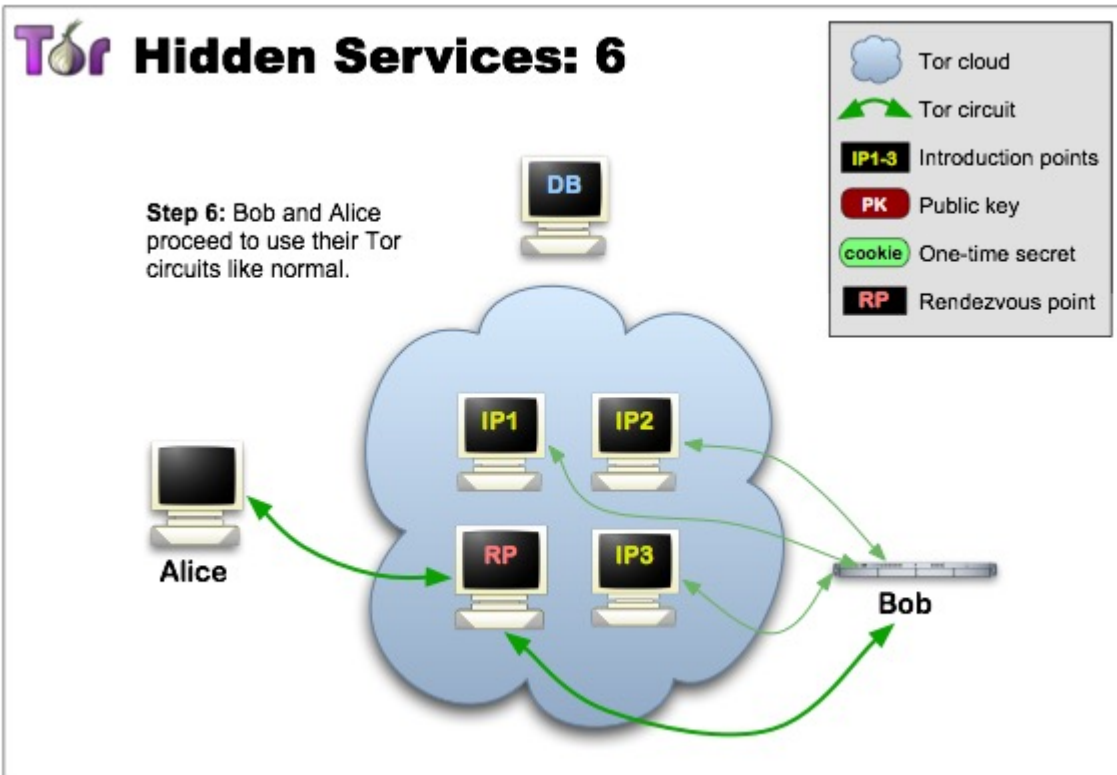
Step 4: Alice writes a message to Bob (encrypted to PK) listing the rendezvous point and a one-time secret, and asks an introduction point to deliver it to Bob.



Tor Hidden Services: 5

Step 5: Bob connects to the Alice's rendezvous point and provides her one-time secret.





- Tor relays is publicly accessible
- Some countries block these IPs
- Tor Bridges
 - Tor proxy, that is not included in the directory
- Tor packets can be detected using the DPI (fix cells)
 - Traffic obfuscating – traffic acts as HTTP, Bittorrent ...

I2P networks

- Encrypted peer-to-peer overlay network
- Not using directory servers (as Tor), but NetDB - self organizing network database
- Packet switched vs circuit switched (Tor)
- Applications build on top of the I2P router sw
 - Email (susimail), IRC,
 - Eepsites
 - I2P equivalent of a Tor Hidden service – site hosted only in I2P network